

Project Documentation

Project: Rising Water – AI-Based Flood Prediction System

1. Project Summary

Rising Water is a Machine Learning-based web application that predicts flood probability using 20 environmental parameters. The system is developed using Python, Flask, XGBoost, HTML, and CSS.

2. Folder Structure

Rising Water/

- └─ app.py
- └─ dataset/
- └─ models/
- └─ templates/
- └─ static/
- └─ requirements.txt
- └─ Project Documentation/

3. Technologies Used

- Python
- Flask
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- HTML
- CSS
- Joblib
- GitHub

4. Key Features

- Flood probability prediction
- Low / Medium / High risk classification
- Responsive web interface
- Fast prediction using trained model
- Easy deployment and maintenance

5. Project Outcome

The project demonstrates the practical use of Machine Learning for disaster preparedness by providing quick and reliable flood risk assessment.

6. Conclusion

The Rising Water project successfully integrates Machine Learning with a Flask web application to provide an easy-to-use flood prediction system.